Chapter 3
Poisson Process

- Exponential distribution
- Poisson process
- Generalization of Poisson process

Exponential distribution

- A random variable X, taking value on $[0, \infty)$ is said to be an exponential random variable with *rate* λ, if its probability density function equals

$$f_X(x) = \lambda e^{-\lambda x}.$$

  Then, its tail

$$P(X \geq x) = e^{-\lambda x}.$$

- $E[X] = \frac{1}{\lambda}$, Var$[X] = \frac{1}{\lambda^2}$

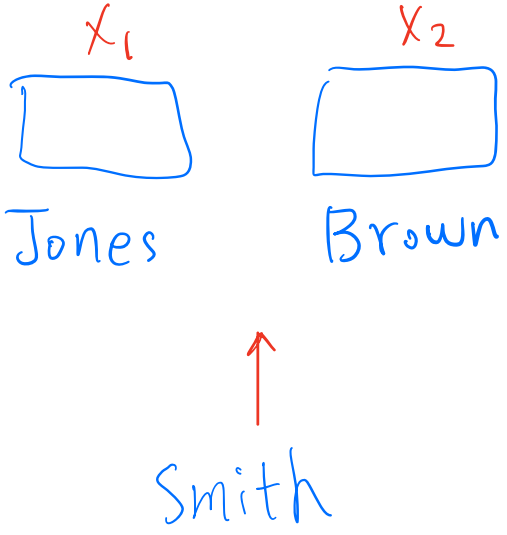
- **Memoryless Property:** For any $a > 0$,

$$P(X > a + x | X > a) = e^{-\lambda x}.$$

  In other words, $X - a$ condition on that $X > a$ is still an exponential r.v. with rate λ.

Memoryless Property

- **Example 5.3** Consider a post office that is run by two clerks. Suppose that when Mr. Smith enters the system he discovers that Mr. Jones is being served by one of the clerks and Mr. Brown by the other. Suppose also that Mr. Smith is told that his service will begin as soon as either Jones or Brown leaves. If the amount of time that a clerk spends with a customer is exponentially distributed with mean $1/\lambda$, what is the probability that, of the three customers, Mr. Smith is the last to leave the post office?



By memoryless property

the chance is $\frac{1}{2}$.

Other properties of exponential

- If X_1 and X_2 are two independent exponential r.v.'s of rate λ_1 and λ_2, then

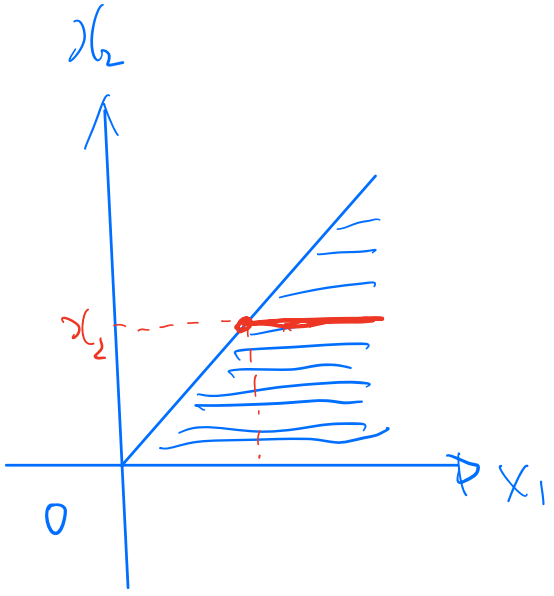
$$P(X_1 > X_2) = \frac{\lambda_2}{\lambda_1 + \lambda_2}.$$

$$= \lambda_1\lambda_2 \int\int_{x_1>x_2} e^{-\lambda_1 x_1} e^{-\lambda_2 x_2} \, dx_1 dx_2$$

$$= \lambda_1\lambda_2 \int_0^\infty \left(\int_{x_2}^\infty e^{-\lambda_1 x_1} \, dx_1 \right) e^{-\lambda_2 x_2} \, dx_2$$

$$= \lambda_2 \int_0^\infty e^{-\lambda_1 x_2 - \lambda_2 x_2} \, dx_2$$

$$= \frac{\lambda_2}{\lambda_1 + \lambda_2}.$$



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In general, let $X_1, \cdots, X_n$ be indept and $X_i \sim \text{Exp}(\lambda_i)$. Then

$$P(X_1 > X_2 > \cdots > X_n)$$

$$= \lambda_1 \cdots \lambda_n \int\cdots\int_{x_n < \cdots < x_2 < x_1} e^{-\lambda_1 x_1} \cdots e^{-\lambda_n x_n} \, dx_1 \cdots dx_n$$

$$= \lambda_1 \cdots \lambda_n \int\cdots\int_{x_n < \cdots < x_2} \left(\int_{x_2}^{\infty} e^{-\lambda_1 x_1} \, dx_1 \right) e^{-\lambda_2 x_2 - \cdots - \lambda_n x_n} \, dx_2 \cdots dx_n$$

$$= \lambda_2 \cdots \lambda_n \int\cdots\int_{x_n < \cdots < x_2} e^{-(\lambda_1 + \lambda_2) x_2 - \lambda_3 x_3 - \cdots - \lambda_n x_n} \, dx_2 \cdots dx_n$$

$$= \frac{\lambda_2 \cdots \lambda_n}{\lambda_1 + \lambda_2} \int\cdots\int_{x_n < \cdots < x_3} e^{-(\lambda_1 + \lambda_2 + \lambda_3) x_3 - \cdots - \lambda_n x_n} \, dx_3 \cdots dx_n$$

$$= \frac{\lambda_2 \cdots \lambda_n}{(\lambda_1 + \lambda_2)(\lambda_1 + \lambda_2 + \lambda_3)} \int\cdots\int_{x_n < \cdots < x_4} e^{-(\lambda_1 + \cdots + \lambda_4) x_4 - \cdots - \lambda_n x_n} \, dx_4 \cdots dx_n$$

$$= \frac{\lambda_2 \lambda_3 \cdots \lambda_n}{(\lambda_1 + \lambda_2)(\lambda_1 + \lambda_2 + \lambda_3)(\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4) \cdots (\lambda_1 + \cdots + \lambda_{n-1})} \int_0^{\infty} e^{-(\lambda_1 + \cdots + \lambda_n) x_n} \, dx_n$$

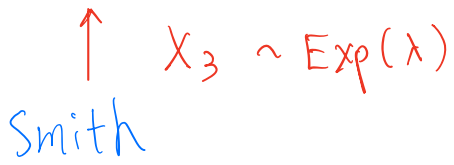
$$= \frac{\lambda_1 \lambda_2 \lambda_3 \cdots \lambda_n}{\lambda_1 (\lambda_1 + \lambda_2) \cdots (\lambda_1 + \cdots + \lambda_n)}.$$

In particular, take $\lambda_1 = \cdots = \lambda_n = \lambda$, we get

$$P(X_1 > X_2 > \cdots > X_n) = \frac{1}{n!}.$$

- **Example 5.3** Consider a post office that is run by two clerks. Suppose that when Mr. Smith enters the system he discovers that Mr. Jones is being served by one of the clerks and Mr. Brown by the other. Suppose also that Mr. Smith is told that his service will begin as soon as either Jones or Brown leaves. If the amount of time that a clerk spends with a customer is exponentially distributed with mean $1/\lambda$, what is the probability that, of the three customers, Mr. Smith is the last to leave the post office?





$$\text{Answer} = P(X_3 > X_1 - X_2 \,,\, X_1 > X_2) + P(X_3' > X_2 - X_1 \,,\, X_2 > X_1)$$

where $X_3 \sim Exp(\lambda_2)$, $X_3' \sim Exp(\lambda_1)$. Now

$$1\text{st probab} = P(X_3 > X_1 - X_2 \mid X_1 > X_2) \cdot P(X_1 > X_2)$$

$$= P(X_3 > X_1) \cdot P(X_1 > X_2)$$

$$= \frac{\lambda_1}{\lambda_1 + \lambda_2} \cdot \frac{\lambda_2}{\lambda_1 + \lambda_2}$$

by memoryless prop. By symmetry, 2nd probab $= \lambda_1\lambda_2/(\lambda_1+\lambda_2)^2$

$$\therefore \text{Answer} = \frac{2\lambda_1\lambda_2}{(\lambda_1+\lambda_2)^2}.$$ In particular, it is $\frac{1}{2}$ if $\lambda_1 = \lambda_2$.

Other properties of exponential

- **Proposition 5.2** Suppose X_1, X_2,...,X_n are independent exponential r.v.'s of rate λ_1, λ_2,...,λ_n respectively. Let

$$Y = \min_{1 \le i \le n} X_i.$$

Then,

$$Y \sim \text{exponential} \left(\sum_{i=1}^{n} \lambda_i \right).$$

Besides, Y is independent of the rank order of $X_1, X_2, ..., X_n$.

In fact, $P(Y \ge t) = P(X_1 > t) \cdots P(X_n > t)$

$= e^{-\lambda_1 t} \cdots e^{-\lambda_n t}$

$= e^{-(\lambda_1 + \cdots + \lambda_n)t} \Rightarrow Y \sim Exp(\sum_{1}^{n} \lambda_i)$

Proof. Let ranks of $X_1, \cdots, X_n$ be $R_1, R_2, \cdots, R_n$, respectively.

Let $(k_1, k_2, \cdots, k_n)$ be a permutation of $\{1, 2, \cdots, n\}$. Then

$$\{R_1 = k_1, R_2 = k_2, \cdots, R_n = k_n\}$$
$$= \{X_i \text{ is the } i\text{th smallest}, 1 \le i \le n\}.$$

For example, $n = 4$.

$$\{R_1 = 3, R_2 = 4, R_3 = 1, R_4 = 2\}$$
$$= \{X_3 < X_4 < X_1 < X_2\}.$$

As a result,

$$P(R_1 = 3, R_2 = 4, R_3 = 1, R_4 = 2 \mid Y \ge t)$$
$$= P(X_3 - t < X_4 - t < X_1 - t < X_2 - t \mid X_1 \ge t, X_2 \ge t, X_3 \ge t, X_4 \ge t)$$

Given $X_i \ge t$, we know $X_i - t \sim Exp(\lambda_i)$, $1 \le i \le 4$. By independence, given $X_i \ge t$, $1 \le i \le 4$, we know $X_i - t \sim Exp(\lambda_i)$, $1 \le i \le 4$.

Thus

$$P(R_1 = 3, R_2 = 4, R_3 = 1, R_4 = 2 \mid Y \ge t)$$
$$= P(X_3 < X_4 < X_1 < X_2)$$
$$= P(R_1 = 3, R_2 = 4, R_3 = 1, R_4 = 2)$$

Obviously, the above holds if $(3, 4, 1, 2)$ is replaced by any permutation of $\{1, 2, 3, 4\}$. Also, it holds if "4" is replaced by "n".

Two Queueing Examples

- **Example 5.8** Suppose you arrive at a post office having two clerks at a moment when both are busy but there is no one else waiting in line. You will enter service when either clerk becomes free. If service times for clerk i are exponential with rate λ_i,$i = 1, 2$, find $E[T]$, where T is the amount of time that you spend in the post office.

Solution. Let $R_i \sim Exp(\lambda_i)$, $i=1,2$, and X_1 and X_2 are indept. $T = \min\{R_1, R_2\} + \tilde{R}_1 1(R_1 < R_2) + \tilde{R}_2 1(R_2 < R_1)$ where $\tilde{R}_1, \tilde{R}_2$ are both indept of R_1, R_2 and $\tilde{R}_i \sim Exp(\lambda_i)$.

Thus $$ET = \frac{1}{\lambda_1+\lambda_2} + \frac{1}{\lambda_1} \cdot \frac{\lambda_1}{\lambda_1+\lambda_2} + \frac{1}{\lambda_2} \cdot \frac{\lambda_2}{\lambda_1+\lambda_2}$$

$$= \frac{3}{\lambda_1+\lambda_2}.$$

Similarly, one can generalize this case to n clerks. Use:

$$T = \min\{R_1, \cdots, R_n\}$$

$$+ \sum_{i=1}^{n} \tilde{R}_i 1(R_i < \min\{R_j, 1 \le j \le n, j \ne i\})$$

$$Y \sim Exp(\lambda_1 + \cdots + \lambda_n - \lambda_i)$$

$$P(R_i < Y) = \frac{\lambda_i}{\lambda_1 + \cdots + \lambda_n}, \text{ where } Y \sim Exp(\lambda_1 + \cdots + \lambda_n - \lambda_i).$$

Thus,

$$ET = \frac{1}{\lambda_1 + \cdots + \lambda_n} + \sum_{i=1}^{n} \frac{1}{\lambda_i} \cdot \frac{\lambda_i}{\lambda_1 + \cdots + \lambda_n} = \frac{n+1}{\lambda_1 + \cdots + \lambda_n}.$$

Eg. Let $X_i, i = 1, \ldots, n$, be independent exponential random variables with respective rates $\lambda_i, i = 1, \ldots, n$, and suppose that $\lambda_i \neq \lambda_j$ for $i \neq j$. The random variable $\sum_{i=1}^n X_i$ is said to be a *hypoexponential* random variable. We compute its probabil-dist. In fact, the pdf of $\sum_{i=1}^n X_i$ is

$$f_{X_1+\cdots+X_n}(t) = \sum_{i=1}^{n} C_{i,n} \lambda_i e^{-\lambda_i t}$$

linear combination of exponential density

where

$$C_{i,n} = \prod_{j \neq i} \frac{\lambda_j}{\lambda_j - \lambda_i}$$

Proof. Look at $n=2$. We know the pdf of $X_1 + X_2$ is

$$f_{X_1+X_2}(t) = \int_0^t f_{X_1}(s) f_{X_2}(t-s)\, ds$$

$$= \int_0^t \lambda_1 e^{-\lambda_1 s} \lambda_2 e^{-\lambda_2 (t-s)}\, ds$$

$$= \lambda_1 \lambda_2 e^{-\lambda_2 t} \int_0^t e^{(\lambda_2 - \lambda_1) s}\, ds$$

$$= \lambda_1 \lambda_2 e^{-\lambda_2 t} \frac{1}{\lambda_2 - \lambda_1} \left[e^{(\lambda_2 - \lambda_1) t} - 1 \right]$$

$$= \lambda_1 \cdot \frac{\lambda_2}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} + \lambda_2 \cdot \frac{\lambda_1}{\lambda_1 - \lambda_2} e^{-\lambda_2 t}. \quad (*)$$

By induction, one can show the conclusion holds for $n \geq 2$.

Q: if $\lambda_1 = \lambda_2$, we know $X_1 + X_2$ is Gamma dist. with pdf

$$g_{X_1+X_2}(t) = \lambda_1^2 t e^{-\lambda_1 t}, \quad t \geq 0.$$

Can we get this from $(*)$?

In fact, from (*),

$$f_{X_1+X_2}(t) = -\lambda_1\lambda_2 \frac{e^{-\lambda_2 t} - e^{-\lambda_1 t}}{\lambda_2 - \lambda_1}.$$

Let $\lambda_2 \to \lambda_1$, we obtain the above has limit

$$-\lambda_1^2 \frac{d}{d\lambda_2} e^{-\lambda_2 t}\Big|_{\lambda_2=\lambda_1} = \lambda_1^2 t e^{-\lambda_1 t}, \quad t \geq 0.$$

Yes, we indeed get it.

In general, if $\lambda_1 = \cdots = \lambda_n$, then the pdf of $X_1 + \cdots + X_n$ is Gamma dist. with pdf

$$\frac{1}{(n-1)!} \lambda_1^n t^{n-1} e^{-\lambda_1 t}, \quad t \geq 0.$$

If interested, you can try to obtain this from

$$f_{X_1+\cdots+X_n}(t) = \sum_{i=1}^{n} C_{i,n} \lambda_i e^{-\lambda_i t}.$$

Proposition 5.3. *Let $X_1, \ldots, X_n$ be independent exponential random variables with rate λ, and let $S = \sum_{i=1}^n X_i$. Then, $(\frac{X_1}{S}, \frac{X_2}{S}, \ldots, \frac{X_{n-1}}{S})$ has a Dirichlet distribution,* i.e., Uniform dist over the set $\{(y_1, \ldots, y_{n-1}) : y_i \geq 0, y_1 + \cdots + y_{n-1} \leq 1\}$.

Proof. From earlier discussion, we know S has pdf

$$f_S(t) = \frac{1}{(n-1)!} \lambda^n t^{n-1} e^{-\lambda t}, \quad t \geq 0.$$

Given $S = t$, if $X_1 = x_1, \ldots, X_{n-1} = x_{n-1}$, then $X_n = t - \sum_1^{n-1} x_i$.

Thus, the conditional pdf of $X_1, \cdots, X_{n-1}$ given $S = t$ is

$$\frac{f_{X_1,\cdots,X_n}(x_1, x_2, \cdots, x_{n-1}, t - \sum_1^{n-1} x_i)}{f_S(t)}$$

$$= \frac{\lambda^n e^{-\lambda x_1} \cdot e^{-\lambda x_2} \cdots e^{-\lambda x_{n-1}} \cdot e^{-(t - \sum_1^{n-1} x_i)\lambda}}{\frac{1}{(n-1)!} \lambda^n t^{n-1} e^{-\lambda t}}$$

$$= (n-1)!\, t^{1-n}$$

for $x_1 \geq 0, \cdots, x_{n-1} \geq 0$. $x_1 + \cdots + x_{n-1} \leq t$. By making a trivial transform, we know the joint pdf of $\frac{X_1}{S}, \cdots, \frac{X_{n-1}}{S}$ given $S = t$ is

$$(n-1)! \text{ on } \{(y_1, \cdots, y_{n-1});\ y_i \geq 0,\ y_1 + \cdots + y_{n-1} \leq 1\}$$

This is a uniform dist, not depending on s, and hence indept of S. Thus the conditional pdf of $\frac{X_1}{S}, \cdots, \frac{X_{n-1}}{S}$ and their unconditional pdf are identical.

Sometimes, for convenience, we simply say

$\left(\frac{X_1}{X_1 + \cdots + X_n}, \cdots, \frac{X_n}{X_1 + \cdots + X_n}\right)$ has uniform dist. on the n-dim simplex: $\{(y_1, \cdots, y_n): y_i \geq 0,\ y_1 + \cdots + y_n = 1\}$.

Set $A_m = \{(y_1, \cdots, y_m) ; y_i \geq 0, y_1 + \cdots + y_m \leq 1\}$. Then

$$\mathrm{Vol}(A_m) = \frac{1}{m!}, \quad m \geq 1.$$

Proof. As $m = 1$, $A_1 = [0, 1]$. So $\mathrm{Vol}(A_1) = 1 = m!$

Suppose the conclusion holds for m, then

$$\mathrm{Vol}(A_{m+1}) = \int\cdots\int_{y_1 + \cdots + y_{m+1} \leq 1} 1 \, dy_1 \cdots dy_{m+1}$$

$$= \int_0^1 dy_{m+1} \int_{\substack{y_1 + \cdots + y_m \leq 1 - y_{m+1} \\ y_i \geq 0}} 1 \, dy_1 \cdots dy_m$$

$x_i = \frac{y_i}{1 - y_{m+1}}$

$$\int_0^1 (1 - y_{m+1})^m \, dy_{m+1} \int_{x_1 + \cdots + x_m \leq 1} 1 \, dx_1 \cdots dx_m$$

$$= \frac{1}{m+1} \cdot \frac{1}{m!}$$

$$= \frac{1}{(m+1)!}.$$

Dirichlet dist. $Y_1, \cdots, Y_{n-1}$ has joint pdf

$$g(y_1, \cdots, y_{n-1}) = \frac{1}{\text{Beta}(\alpha_1, \cdots, \alpha_n)} y_1^{\alpha_1 - 1} \cdots y_{n-1}^{\alpha_{n-1} - 1} (1 - y_1 - \cdots - y_{n-1})^{\alpha_n - 1}$$

for $y_i \geq 0$ and $\sum_{i=1}^{n-1} y_i \leq 1$, where parameter $\alpha_1 > 0, \cdots, \alpha_n > 0$ and

$$\text{Beta}(\alpha_1, \cdots, \alpha_n) = \frac{\Gamma(\alpha_1) \cdots \Gamma(\alpha_n)}{\Gamma(\alpha_1 + \cdots + \alpha_n)}.$$

Remarks. 1. Dirichlet dist. is a generalization of Beta dist.

2. Our example above corresponds to $\alpha_i \equiv 1$.

3. There is an analogue of the Dirichlet dist. with $n = \infty$. This case is called Poisson-Dirichlet dist.

A ~~Two~~ Queueing Example~~s~~

- **Example 5.10** Suppose that customers are in line to receive service that is provided sequentially by a server; whenever a service is completed, the next person in line enters the service facility. However, each waiting customer will only wait an exponentially distributed time with rate θ; if its service has not yet begun by this time then it will immediately depart the system. These exponential times, one for each waiting customer, are independent. In addition, the service times are independent exponential random variables with rate μ. Suppose that someone is presently being served and consider the person who is n-th in line.

~~(a) Find P_n, the probability that this customer is eventually served.~~

~~(b) Find W_n, the conditional expected amount of time this person spends waiting in line given that she is eventually served.~~

Fin Q_n, the probability that none of the n customers leave.

Solution. Let (X_i, Y_i), $i = 1, 2, \cdots, n$ be iid with dist $X_i \sim \text{Exp}(\mu)$ and $Y_i \sim \text{Exp}(\theta)$ and X_i and Y_i are indept.

We use X_i and Y_i to represent the service time and waiting time for each i.

Service time — waiting time

1st (X_1, Y_1)

2nd (X_2, Y_2), $\quad Y_2 > X_1$

3rd (X_3, Y_3) $\quad Y_3 > X_2 + X_1$

4th (X_4, Y_4) $\quad Y_4 > X_2 + X_1 + X_3$

$$\therefore \; P(Y_2 > X_1,\ Y_3 > X_1 + X_2,\ Y_4 > X_1 + X_1 + X_3)$$

$$\underset{\text{on } X_1, X_2, X_3}{\overset{\text{Conditioning}}{=}} E\, e^{-\theta X_1} e^{-\theta(X_1 + X_2)} \cdot e^{-\theta(X_1 + X_2 + X_3)}$$

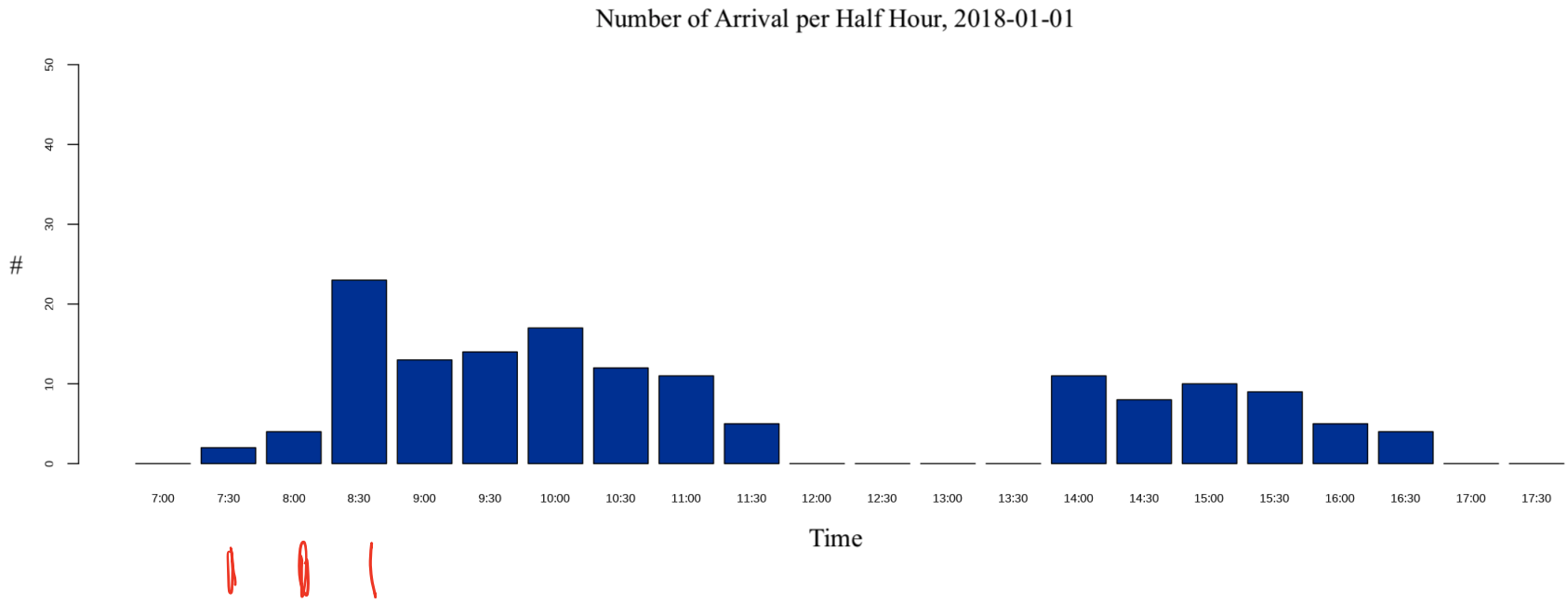
$$= E\, e^{-3\theta X_1} e^{-2\theta X_2} e^{-\theta X_3} = \frac{\mu}{\mu + 3\theta} \frac{\mu}{\mu + 2\theta} \frac{\mu}{\mu + \theta}$$

$$\therefore \; P(Y_2 > X_1,\ Y_3 > X_1 + X_2,\ \cdots,\ Y_n > X_1 + \cdots + X_{n-1})$$

$$= \frac{\mu^{n-1}}{(\mu + \theta)(\mu + 2\theta) \cdots (\mu + (n-1)\theta)}.$$

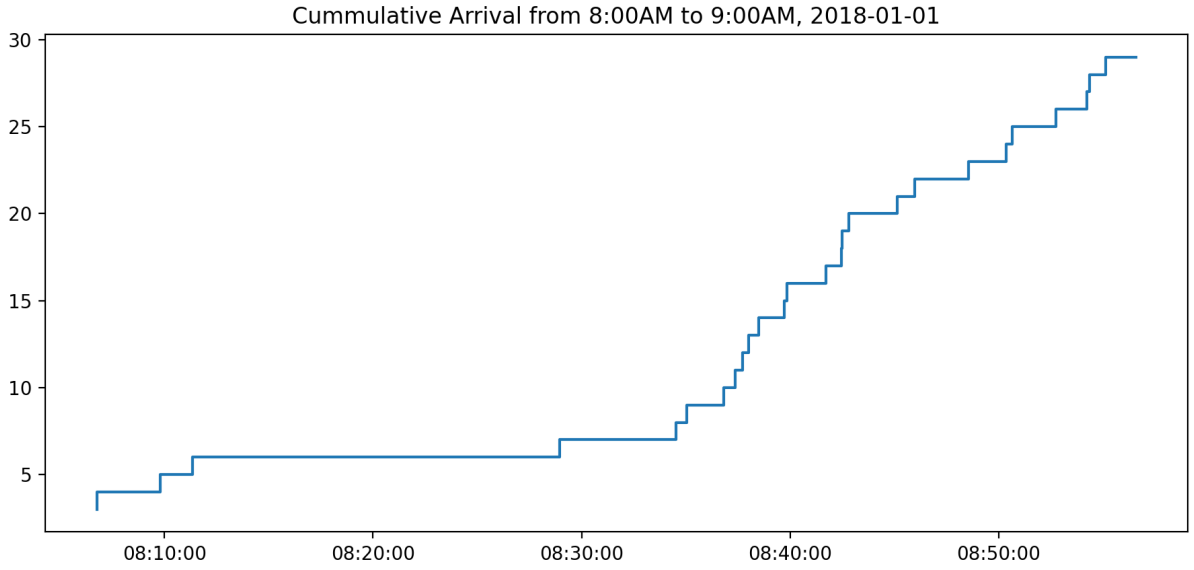
Modeling Patient Arrivals

- Patient arrivals to ultrasound center in a hospital.



Modeling Patient Arrivals

- Patient arrivals to ultrasound center in a hospital.



Counting Process

- A continuous-time stochastic process $\{N(t) : t \geq 0\}$ is called a counting process if
  1. $N(t) \geq 0$.
  2. $N(t)$ is integer valued.
  3. If $s < t$, then $N(s) \leq N(t)$.
- We say a counting process $\{N(t) : t \geq 0\}$ has independent increments if for any finite ~~pairs of time points $s_1 < t_1 < s_2 < t_2 < \cdots$~~, the random variables

  $$N(t_1) - N(s_1), N(t_2) - N(s_2), ..., N(t_n) - N(s_n)$$

  are independent of each other.

disjoint intervals $[s_i, t_i]$

- We say a counting process $\{N(t) : t \geq 0\}$ has stationary increments if for any $s, t > 0$, the two random variables

  $$N(t) \text{ and } N(t+s) - N(s)$$

  have the same distribution.

Poisson Process

Definition 5.2. The counting process $\{N(t), t \geq 0\}$ is said to be a Poisson process with rate $\lambda > 0$ if the following axioms hold:

(i) $N(0) = 0$

(ii) $\{N(t), t \geq 0\}$ has independent increments

(iii) $P(N(t+h) - N(t) = 1) = \lambda h + o(h)$

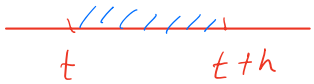
(iv) $P(N(t+h) - N(t) \geq 2) = o(h)$

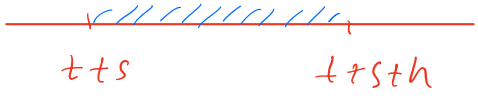
(iii) and (iv) depend on increments, not locations

(ii) implies $N(t+s) - N(s)$ is indept of $N(s_1), \cdots, N(s_m)$, $s_i \leq s$.

- **Lemma 5.1** For any $s \geq 0$, define $N_s(t) \triangleq N(t+s) - N(s)$ for all $t \geq 0$. Then $\{N_s(t) : t \geq 0\}$ is also a Poisson process with rate λ.

$N_s(t+h) - N_s(t) = N(t+s+h) - N(t+s)$



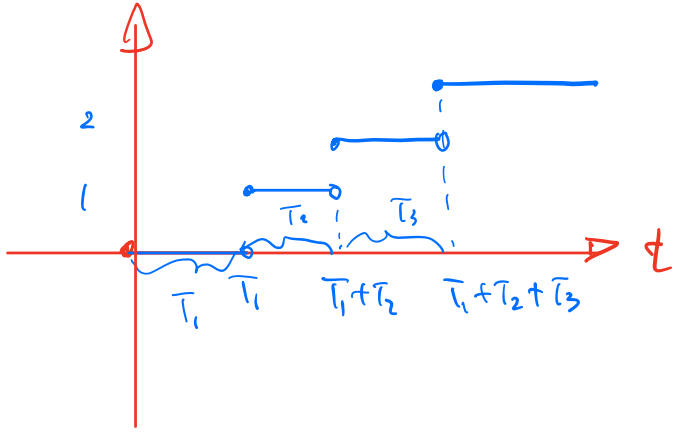


Poisson Process

- **Lemma 5.2** Let T_1 be the time of the first arrival of the Poisson process $\{N(t) : t \geq 0\}$, then

$$P(T_1 > t) = P(N(t) = 0) = e^{-\lambda t}.$$

$\{T_1 > t\} = \{N(t) = 0\}$



Proof. Set $P_0(t) = P(N(t)=0)$, $t \geq 0$.

obviously, $P_0(0) = 1$. By def.

$$P_0(t+h) = P(N(t+h)=0)$$

$$= P(N(t)=0,\ N(t+h)-N(t)=0)$$

for $h \geq 0$. Use indept. increments to see

$$P_0(t+h) = P(N(t)=0) \cdot P(N(t+h)-N(t)=0)$$

$$= P_0(t) \cdot (1-\lambda h + o(h)) \qquad (*)$$

since (iii) + (iv) implies

$$P(N(t+h)-N(t)=0) = 1-(\lambda h + o(h)) - o(h)$$

$$= 1-\lambda h + o(h).$$

Again this implies

$$\frac{P_0(t+h)-P_0(t)}{h} = -\lambda P_0(t) + o(1)$$

for $h > 0$. Let $h \downarrow 0$ and we see

$$P_0'(t) = -\lambda P_0(t), \quad t \geq 0. \qquad (**)$$

This is equivalent to

$$\left[e^{\lambda t} P_0(t)\right]' = 0$$

$(fg)' = f'g + fg'$

Why? $LHS = \lambda e^{\lambda t} P_0(t) + e^{\lambda t} P_0'(t)$

$= e^{\lambda t}\left(\lambda P_0(t) + P_0'(t)\right)$

$= 0$

by (**). $\therefore e^{\lambda t} P_0(t) \equiv c, \quad t \geq 0.$

Take $t=0$ and use $P_0(0)=1$ to see $c=1$.

Thus, $P_0(t) = e^{-\lambda t}$.



Interarrival Times

- Let T_n be the time between the $n-1$-th and n-th arrivals. The sequence $\{T_n\}$ is called the sequence of interaarival times.

- **Proposition 5.4** $T_1, T_2, ...$ are independent and identically distributed exponential random variables with rate λ.

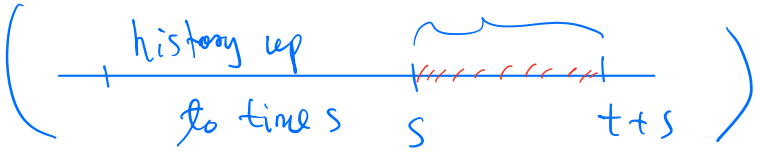
Proof. We will show

$$P(T_2 > t \mid T_1 = s) = e^{-\lambda t} \qquad (*)$$

for any $s > 0$, $t > 0$. This implies T_1 & T_2 are indept. and $P(T_2 > t) = P(T_1 > t) = e^{-\lambda t}$.

Thus, T_1 and T_2 are indept and each has dist. of $Exp(\lambda)$. In fact, the LHS of $(*)$ is

$$P\big(N(t+s) - N(s) = 0 \mid N(s) = 1, \quad N(s') < 1 \text{ for any } s' < s\big)$$



$$= P\big(\underbrace{N(t+s) - N(s)}_{N_s(t)} = 0\big)$$

$$= e^{-\lambda t}$$

by Lemma 5.1. We obtain $(*)$.

Similarly, consider

$$P(T_3 > t \mid T_1 = s_1, T_2 = s_2)$$

$$= P\big(N(t + s_1 + s_2) - N(s_1 + s_2) = 0 \mid N(s_1) = 1, N(s') = 0 \text{ for } 0 < s' < s_1, N(s_1 + s_2) = 2 \text{ and } N(s'') = 1 \text{ for } s_1 < s'' < s_1 + s_2\big)$$

messy

history up to S_1+S_2.

$$= P\left(N_{S_1+S_2}(t) = 0\right)$$

$$= e^{-\lambda t}$$

Therefore, T_3 is indept of T_1 & T_2 and $T_3 \sim Exp(\lambda)$.

Similarly, we see $T_1, T_2, T_3, \ldots$ are iid $Exp(\lambda)$.

Increment Distribution

- **Theorem 5.1** If $\{N(t) : t \geq 0\}$ is a Poisson process with rate λ, then for any $t > 0$, $N(t)$ is a Poisson random variable with mean λt.

Review: X is a r.v. g is a fun. Y is a r.v. Then $E\,g(X) = E\left[E(g(X)\,|\,Y)\right]$. For any set A, take $g(x) = 1(x \in A)$. Then $P(X \in A) = E\,P(X \in A\,|\,Y)$.

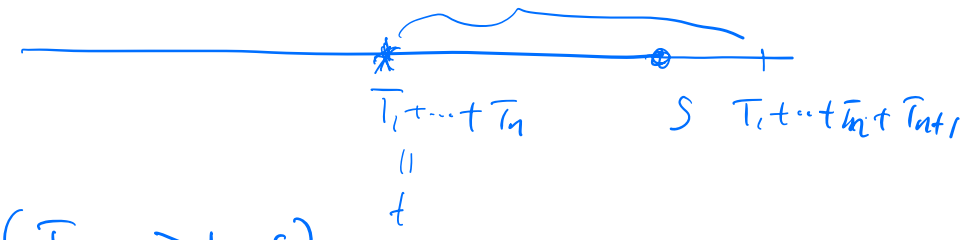
Proof. As seen earlier, T_n is the time between $(n-1)$-th and n-th arrival. Then $T_1, T_2, T_3, \cdots, T_n$ are iid $\text{Exp}(\lambda)$-dist'd r.v.'s. Thus $T := T_1 + \cdots + T_n$ follows the Gamma dist. with pdf

$$f_n(t) = \frac{\lambda^n t^{n-1}}{(n-1)!} e^{-\lambda t}, \quad t \geq 0. \qquad (*)$$

Observe, for $t > s$,

$$P(N(t) = n \mid T = s)$$

$$= P(T_{n+1} > t - s \mid T_1 + T_2 + \cdots + T_n = s)$$



$$\overset{\substack{T_1, T_2, \ldots \\ \text{are iid}}}{=} P(T_{n+1} > t - s)$$

$$= e^{-\lambda(t-s)}.$$

$$\therefore \quad P(N(t) = n \mid T) = e^{-\lambda(t-T)} 1(T < t)$$

Now

$$P(N(t) = n) = E\, P(N(t) = n \mid T)$$

$$= E\, e^{-\lambda(t-T)} 1(T < t)$$

$$\therefore P(N(t)=n)$$

$$\overset{(*)}{=} \int_{-\infty}^{\infty} e^{-\lambda(t-s)} \underbrace{f_n(s)}_{\text{pdf of } T} ds$$

$$= \int_{-\infty}^{\infty} e^{-\lambda(t-s)} \frac{\lambda^n s^{n-1}}{(n-1)!} e^{-\lambda s} \mathbb{1}(0 \le s \le t)\, ds$$

$$= \underbrace{\int_0^t s^{n-1} ds}_{\frac{1}{n}t^n} \cdot \frac{\lambda^n e^{-\lambda t}}{(n-1)!}$$

$$= \frac{\lambda^n t^n}{n!} e^{-\lambda t}.$$



Increment Distribution

- **Example 5.13** Suppose that people immigrate into a territory according to a Poisson process with rate $\lambda = 2$ per day.

  (a) Find the probability there are 10 arrivals in the following week (of 7 days).

  (b) Find the expected number of days until there have been 20 arrivals.

  (c) Find the probability that, in the following week, there are 8 arrivals in the weekdays and 2 arrivals on the weekend.

Solution: Let $N(t)$ be a Poisson process with rate $\lambda = 2$.

(i) Use $N(t) \sim \text{Poi}(\lambda t)$ to see

$$P(N(7) = 10) = \frac{14^{10}}{10!} e^{-14}.$$

(ii) Let $T_1, T_2, \cdots$ be the interarrival time. Then they are iid $\text{Exp}(\lambda)$. Then

$$E(T_1 + \cdots + T_{20}) = 20 \cdot E T_1 = \frac{20}{2} = 10.$$

(iii) $P(N(5) = 8, N(7) - N(5) = 2)$

$$= P(N(5) = 8) \cdot P(N(2) = 2)$$

$$= \frac{10^8}{8!} e^{-10} \frac{4^2}{2!} e^{-4}.$$

Poisson Thinning Theorem

- Consider a Poisson process $\{N(t), t \geq 0\}$ having rate λ, and suppose that each time an event occurs it is classified as either a type I or a type II event. Suppose further that each event is classified as a type I event with probability p or a type II event with probability $1 - p$, independently of all other events.
- Let $N_1(t)$ and $N_2(t)$ denote respectively the number of type I and type II events occurring in $[0, t]$.
- **Proposition 5.5** $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ are two independent Poisson processes having respective rates λp and $\lambda(1 - p)$.
- The result can be generalized so that there are k types and each event is independently classified as type i with probability p_i for $1 \leq i \leq k$.

Proof 1st, $N_1(t) = 0$. Easily, $N_1(t)$ has indep increments. By the def of $N(t)$ being a Poisson process,

(a) $P(N_1(t+h) - N_1(t) = 1)$

$= P(N_1(t+h) - N_1(t) = 1 \mid N(t+h) - N(t) = 1) \cdot P(N(t+h) - N(t) = 1)$

$+ P(N_1(t+h) - N_1(t) = 1 \mid N(t+h) - N(t) \geq 2) \cdot P(N(t+h) - N(t) \geq 2)$

$= p \cdot (\lambda h + o(h)) + o(h)$

$= (\lambda p) h + o(h)$.

(b) $P(N_1(t+h) - N_1(t) \geq 2)$

$\leq P(N(t+h) - N(t) \geq 2) = o(h)$

since $N(t+h) - N(h) = [N_1(t+h) - N_1(t)] + [N_2(t+h) - N_2(t)]$.

$\therefore N_1(t)$ is a Poisson process with rate λp,

Similarly $N_2(t)$ - - - — — — - - - $\lambda(1-p)$.

Now $P(N_1(t) = m, N_2(t) = n)$

$= P(N_1(t) = m \mid N(t) = m+n) \cdot P(N(t) = m+n)$

$= \binom{m+n}{m} p^m q^n \cdot \frac{(\lambda t)^{m+n}}{(m+n)!} e^{-\lambda t}$

$= \frac{(\lambda p t)^m}{m!} e^{-\lambda p t} \cdot \frac{(\lambda q t)^n}{n!} e^{-\lambda q t}$

$= P(N_1(t) = m) \cdot P(N_2(t) = n)$.

for any $m \geq 0$, $n \geq 0$.

Examples

- **Example 5.17 (The Coupon Collecting Problem)** There are m different types of coupons. Each time a person collects a coupon it is, independently of ones previously obtained, a type j coupon with probability p_j , $\sum_{j=1}^{m} p_j = 1$. Let N denote the number of coupons one needs to collect in order to have a complete collection of at least one of each type. Find $E[N]$.

Solution. Modelling:

Suppose the process of the occurrence of coupons is Poisson. According to the story, once the coupon occurs, it is of type j with probab. p_j. Then

$N_j(t)$ = the process of the occurrence of type j coupon is Poisson with rate $p_j\lambda$, $j = 1, 2, \cdots, m$.

According to an earlier thm, $N_1(t), N_2(t), \cdots, N_m(t)$ are indept. Then X_j, the time the 1st type j coupon occurs, are indept and $X_j \sim \text{Exp}(\lambda p_j)$.

We mean to compute $E \max_{1\le j\le n} X_j$.

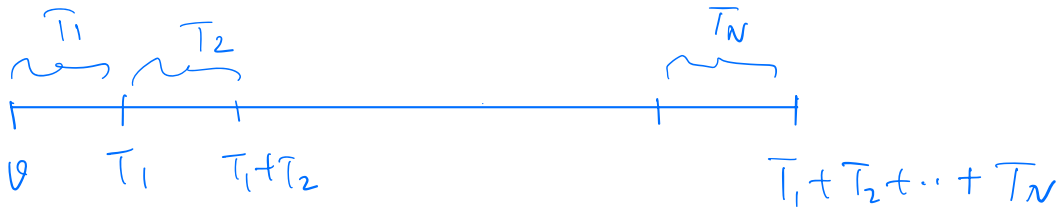
1st, $$P\left(\max_{1\le j\le n} X_j \ge t\right) = 1 - \prod_{j=1}^{n} P(X_j \le t)$$

$$= 1 - \prod_{j=1}^{n} (1 - e^{-\lambda p_j t}).$$

Thus $$E \max_{1\le j\le n} X_j = \int_0^\infty P\left(\max_{1\le j\le n} X_j \ge t\right) dt$$

$$= \int_0^\infty \left[1 - \prod_{j=1}^{n} (1 - e^{-\lambda p_j t})\right] dt.$$

Let T_i = interarrive time between ith and $(i+1)$-th events.



Let N be the number of coupons the collector has when he firstly gets the full package. Then N is indept of T_i's and T_i are iid $Exp(\lambda)$. Observe

$$\max_{1\le j\le n} X_j = \sum_{i=1}^{N} T_i$$

$$\therefore \quad E\max_{1\le j\le n} X_j = ET_1 \cdot EN = \frac{1}{\lambda} EN.$$

$$\therefore \quad EN = \lambda \cdot \int_0^\infty \left[1 - \prod_{j=1}^{n} (1-e^{-\lambda p_j t})\right] dt.$$

Arrival Times of Different Events

- Let $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ be two independent Poisson processes having respective rates λ_1 and λ_2.
- Let $S_1(n)$ denote the time of the n-th event of the first process,and $S_2(m)$ the time of the m-th event of the second process.
- What is $P(S_1(n) < S_2(m))$?

Fact. Flip a coin repeatedly. $P(\text{Heads}) = p$, $P(\text{Tails}) = q$. What is the chance that n Heads occur before m tails?

1st, let's classify the event

$\{n$ H occur before m T$\}$

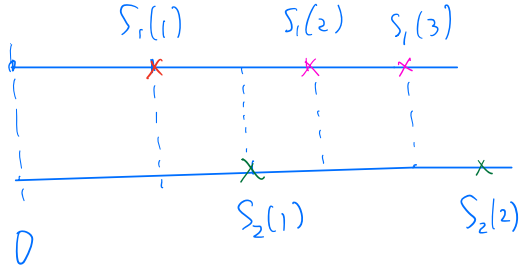
$$= \bigcup_{j=0}^{m-1} \underbrace{\square\square\cdots\square}_{n-1 \text{ Heads and } j \text{ tails}} H$$

Thus

$$\text{the probab.} = \sum_{j=0}^{m-1} \binom{n+j-1}{n-1} p^n q^j = \sum_{k=n}^{n+m-1} \binom{n+m-1}{k} p^k q^{n+m-k-1}$$

Back to the problem, we know $S_1(1)$ and $S_2(1)$ are independent $\text{Exp}(\lambda_1)$ and $\text{Exp}(\lambda_2)$, respectively. Hence

$$P(S_1(1) < S_2(1)) = \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$



After $S_1(1)$, recall

$N_i(t+s) - N_i(s) \overset{d}{=} N_i(t)$

$i = 1, 2.$

Similar to coin flipping, $p = \frac{\lambda_1}{\lambda_1+\lambda_2}$, $q = \frac{\lambda_2}{\lambda_1+\lambda_2}$. Then

$$P(S_1(n) < S_2(m)) = \sum_{j=0}^{m-1} \binom{n+j-1}{n-1} \left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^n \left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^j.$$

$$= \sum_{k=n}^{m+n-1} \binom{m+n-1}{k} \left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^{m+n-k-1}.$$

Let $X_1, X_2, \cdots$ be iid Ber(p), i.e., $P(X_1=1)=p$, $P(X_1=0)=q$. And $X_1+\cdots+X_n \sim$ Bin(n, p), we know

$$P(X_1+\cdots+X_n=k)=\binom{n}{k}p^k q^{n-k}.$$

Thus

$$\sum_{k=n}^{m+n-1}\binom{m+n-1}{k}p^k q^{m+n-1-k}$$

$$= P(X_1+\cdots+X_{m+n-1}\geq n)$$

$$= P\left(\frac{X_1+\cdots+X_{m+n-1}-(m+n-1)p}{\sqrt{(m+n-1)pq}} \geq \frac{n-(m+n-1)p}{\sqrt{(m+n-1)pq}}\right)$$

$$\simeq 1-\Phi\left(\frac{n-(m+n-1)p}{\sqrt{(m+n-1)pq}}\right), \text{ where } \Phi(x)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{x} e^{-t^2/2}dt,$$

by the central limit thm. In particular, if $p=\frac{1}{2}$, $m=n$, the above is approximately $\frac{1}{2}$. In fact, check directly, in this case, the probab. is exactly $\frac{1}{2}$.

We now justify

$$\sum_{j=0}^{m-1} \binom{n+j-1}{n-1} p^n q^j = \sum_{k=n}^{m+n-1} \binom{m+n-1}{k} p^k q^{m+n-k-1}.$$

In fact, define

$$f_n(x) = \sum_{j=0}^{m-1} \binom{n+j-1}{j} x^j,$$

$$g_n(x) = \sum_{i=0}^{m-1} \binom{n+m-1}{n+i} (1-x)^i x^{m-i-1}, \quad x \in [0,1].$$

Claim: $f_n(x) = g_n(x)$, $x \in [0,1]$ (*)

Fact: $\binom{a}{b} - \binom{a-1}{b} = \binom{a-1}{b-1}$, $a \geq b$ (**)

Proof. $$LHS = \frac{a!}{b!(a-b)!} - \frac{(a-1)!}{b!(a-b-1)!}$$

$$= \frac{(a-1)!}{b!(a-b)!}\left[a - (a-b)\right]$$

$$= \binom{a-1}{b-1}$$

Proof. If $n=1$, $f_n(x) = \sum_{j=0}^{m-1} x^j = \frac{1-x^m}{1-x}$. And

$$g_n(x) = \sum_{i=0}^{m-1} \binom{m}{i+1} (1-x)^i x^{m-i-1}$$

$$\overset{j=i+1}{=} \sum_{j=1}^{m} \binom{m}{j} (1-x)^{j-1} x^{m-j}$$

$$= \frac{1}{1-x}\left[\sum_{j=0}^{m} \binom{m}{j} (1-x)^j x^{m-j} - x^m\right]$$

$$= \frac{1}{1-x}(1-x^m).$$

Now we prove (*) by induction.

Assume (*) holds. We only need to check

$$f_{n+1}(x) - f_n(x) = g_{n+1}(x) - g_n(x).$$

In fact,

$$f_{n+1}(x) - f_n(x) = \sum_{j=0}^{m-1} \left[\binom{n+j}{j} - \binom{n+j-1}{j} \right] x^j$$

$$\overset{(**)}{=} \sum_{j=1}^{m-1} \binom{n+j-1}{j-1} x^j$$

$$= \sum_{j=1}^{m-1} \binom{n+j-1}{n} x^j. \qquad (\star)$$

Now $g_{n+1}(x) - g_n(x)$

$$= \sum_{i=0}^{m-2} \left[\binom{n+m}{m-i-1} - \binom{n+m-1}{m-i-1} \right] (1-x)^i x^{m-i-1}$$

$$\overset{(**)}{=} \sum_{i=0}^{m-2} \binom{n+m-1}{m-i-2} (1-x)^i x^{m-i-1}$$

which is again equal to

$$\sum_{i=0}^{m-2} \binom{n+m-1}{n+i+1} (1-x)^i x^{m-i-1}$$

$i+1=j$

$$\sum_{j=1}^{m-1} \binom{n+m-1}{n+j} (1-x)^{j-1} x^{m-j}$$

$$= \frac{x}{1-x} \sum_{j=1}^{m-1} \binom{n+m-1}{n+j} (1-x)^j x^{m-j-1}$$

$$= \frac{x}{1-x} \left[g_n(x) - \binom{n+m-1}{n} x^{m-1} \right]$$

by induction

$$\frac{x}{1-x} \left[\sum_{j=0}^{m-1} \binom{n+j-1}{j} x^j - \binom{n+m-1}{m-1} x^{m-1} \right]$$

$$= \frac{x}{1-x} \sum_{j=0}^{m-2} \binom{n+j-1}{j} x^j + \frac{x}{1-x} \left[\binom{n+m-2}{m-1} - \binom{n+m-1}{m-1} \right] x^{m-1}$$

$$= \frac{x}{1-x} \sum_{j=0}^{m-2} \binom{n+j-1}{j} x^j - \frac{x}{1-x} \binom{n+m-2}{m-2} x^{m-1}.$$

Compare this with (*), we need to verify

$$(1-x) \sum_{j=1}^{m-1} \binom{n+j-1}{n} x^j$$

$$= \left[\sum_{j=0}^{m-2} \binom{n+j-1}{j} x^{j+1} \right] - \binom{n+m-2}{m-2} x^m$$

In fact,

$$\text{LHS} = \sum_{j=1}^{m-1} \binom{n+j-1}{n} x^j - \sum_{j=1}^{m-1} \binom{n+j-1}{n} x^{j+1}$$

$$= \sum_{j=1}^{m-1} \binom{n+j-1}{n} x^j - \sum_{i=2}^{m} \binom{n+i-2}{n} x^i$$

(by setting $j+1=i$)

$$= x - \binom{n+m-2}{n} x^m + \sum_{j=2}^{m-1} \left[\binom{n+j-1}{n} - \binom{n+j-2}{n} \right] x^j$$

$$= x - \binom{n+m-2}{n} x^m + \sum_{j=2}^{m-1} \binom{n+j-2}{n-1} x^j$$

$$= -\binom{n+m-2}{n} x^m + \sum_{j=1}^{m-1} \binom{n+j-2}{n-1} x^j$$

$$\overset{j-1=i}{=} -\binom{n+m-2}{n} x^m + \sum_{i=0}^{m-2} \binom{n+i-1}{n-1} x^{i+1}$$

$$= -\binom{n+m-2}{m-2} x^m + \sum_{j=0}^{m-2} \binom{n+j-1}{j} x^{j+1}.$$

Conditional Distribution of Arrival Times

- **Theorem 5.2** Given that $N(t) = n$, the n arrival times $S_1, ..., S_n$ have the same distribution as the *order statistics* corresponding to n independent random variables uniformly distributed on the interval $(0, t)$.

- **Application:** Sampling a Poisson process on $[0, t]$. Two Ways

  1. Simulate interarrival times $T_1, T_2, ...$ until $\sum_{i=1}^{n} T_i > t$. Output $T_1, ..., T_{n-1}$.
  2. Simulate a Poisson random variable N with mean λt. Then, generate N i.i.d. uniform random variables on $(0, t)$ and output the order statistics.

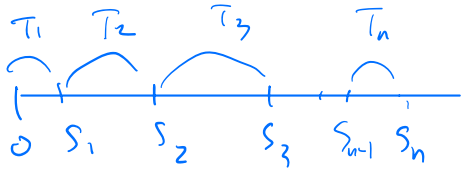
1st, Let $X_1, X_2, \cdots, X_n$ be iid $U[0,t]$.



Then the order statistics $(X_{(1)}, X_{(2)}, \cdots, X_{(n)})$ has pdf $n! f(x_1)\cdots f(x_n)$, where $f(x) = \frac{1}{t}$.

Thus, $(X_{(1)}, \cdots, X_{(n)})$ has pdf $\frac{n!}{t^n}$

<u>Solution</u>. Let $A \subset \mathbb{R}^n$. Then



$$P\left((S_1, S_2, \cdots, S_n) \in A \mid N(t)=n\right)$$

$$= \frac{P\left((T_1, T_1+T_2, \cdots, T_1+\cdots+T_n) \in A, \overbrace{T_1+\cdots+T_n<t}^{} \; T_1+\cdots+T_{n+1}>t\right)}{P(N(t)=n)}$$

$$= \int_{\substack{(t_1, t_1+t_2, \cdots, t_1+\cdots+t_n)\in A, \\ t_{n+1} > t-(t_1+\cdots+t_n) \\ t_1+\cdots+t_n<t}} \frac{\lambda e^{-\lambda t_1}\lambda e^{-\lambda t_2}\cdots\lambda e^{-\lambda t_n}\lambda e^{-\lambda t_{n+1}}}{\frac{(\lambda t)^n}{n!}e^{-\lambda t}}\, dt_1\cdots dt_{n+1}$$

$$= \int_{\substack{(t_1, t_1+t_2, \cdots, t_1+\cdots+t_n)\in A, \\ t_{n+1} > t-(t_1+\cdots+t_n)}} \frac{\lambda e^{-\lambda t_1}\cdot\lambda e^{-\lambda t_2}\cdots\lambda e^{-\lambda t_n}\cdot\lambda e^{-\lambda t_{n+1}}}{\frac{(\lambda t)^n}{n!}e^{-\lambda t}}\, dt_1\cdots dt_{n+1}$$

$$= \int_{(t_1, t_1+t_2, \cdots; t_1+\cdots+t_n) \in A} \frac{n!}{t^n}\, dt_1 \cdots dt_n$$

That is

$$P((S_1, \cdots, S_n) \in A \mid N(t) = n) = \int_{(S_1, \cdots, S_n) \in A} \frac{n!}{t^n}\, ds_1 \cdots ds_n$$

by changing variable $S_i = t_1 + \cdots + t_i$.

Thus, given $N(t) = n$, the pdf of $(S_1, \cdots, S_n)$ is a const, thus a uniform dist. on

$$\{(S_1, \cdots, S_n) \in [0,t]^n;\ 0 \le S_1 < S_2 < \cdots < S_n \le t\}.$$

Three equivalent definitions of Poisson processes

(Leave one as a hw problem)

Recall the def. of Poisson process

Definition 5.2. The counting process $\{N(t), t \geq 0\}$ is said to be a Poisson process with rate $\lambda > 0$ if the following axioms hold:

(i) $N(0) = 0$

(ii) $\{N(t), t \geq 0\}$ has independent increments

(iii) $P(N(t+h) - N(t) = 1) = \lambda h + o(h)$

(iv) $P(N(t+h) - N(t) \geq 2) = o(h)$

There are 2 other equivalent conditions.

Def 5.2' A Poisson process $N(t)$ with rate λ is a stationary process with independent increments and $N(0) = 0$, $N(t) \sim \text{Poi}(\lambda t)$.

Def 5.2'' Let $\xi_1, \xi_2, \cdots$ be iid $\text{Exp}(\lambda)$ and $S_0 = 0$, $S_n = \xi_1 + \cdots + \xi_n$, $n \geq 1$. Then

$$N(t) = \max\{k ; S_k \leq t\}$$

is a Poisson process. The above is also called a renewal process.

From Def 5.2', it is easy to see that, if $N_1(t), \cdots, N_k(t)$ are indept Poisson processes with rate $\lambda_1, \lambda_2, \cdots, \lambda_k$. Then $N_1(t) + \cdots + N_k(t)$ is a Poisson process with rate $\lambda_1 + \cdots + \lambda_k$.

Poisson processes as renewal processes

Reference only

The Poisson process is a special case of renewal process in which the interrenewal times are exponentially distributed. Namely, let $\xi_1, \xi_2, \ldots$ be independent identically distributed random variables having exponential distribution with parameter $\lambda > 0$, that is

$$\mathbb{P}[\xi_k \le x] = 1 - e^{-\lambda x}, \quad x \ge 0.$$

Define the renewal times S_n by

$$S_n = \xi_1 + \ldots + \xi_n, \quad n \in \mathbb{N}, \quad S_0 = 0.$$

It's an exercise to show (for example, by induction) that the density of S_n is given by

$$f_{S_n}(x) = \frac{\lambda^n x^{n-1}}{(n-1)!} e^{-\lambda x}, \quad x \ge 0.$$

The distribution of S_n is called the Erlang distribution with parameters n and λ. It is a particular case of the Gamma distribution.

DEFINITION 3.3.1. The Poisson process with intensity $\lambda > 0$ is a process $\{N_t : t \ge 0\}$ defined by

$$N_t = \sum_{k=1}^{\infty} \mathbb{1}_{\{S_k \le t\}}.$$

Note that N_t counts the number of renewals in the interval $[0, t]$. The next theorem explains why the Poisson process was named after Poisson.

THEOREM 3.3.2. *For all $t \ge 0$ it holds that $N_t \sim \mathrm{Poi}(\lambda t)$.*

PROOF. We need to prove that for all $n \in \mathbb{N}_0$,

$$\mathbb{P}[N_t = n] = \frac{(\lambda t)^n}{n!} e^{-\lambda t}.$$

STEP 1. Let first $n = 0$. Then,

$$\mathbb{P}[N_t = 0] = \mathbb{P}[\xi_1 > t] = e^{-\lambda t},$$

thus establishing the required formula for $n = 0$.

STEP 2. Let $n \in \mathbb{N}$. We compute the probability $\mathbb{P}[N_t = n]$. By definition of N_t we have

$$\mathbb{P}[N_t = n] = \mathbb{P}[N_t \ge n] - \mathbb{P}[N_t \ge n+1] = \mathbb{P}[S_n \le t] - \mathbb{P}[S_{n+1} \le t].$$

Using the formula for the density of S_n we obtain that

$$\mathbb{P}[N_t = n] = \int_0^t f_{S_n}(x)dx - \int_0^t f_{S_{n+1}}(x)dx = \int_0^t \left(\frac{\lambda^n x^{n-1}}{(n-1)!} e^{-\lambda x} - \frac{\lambda^{n+1} x^n}{n!} e^{-\lambda x} \right) dx.$$

The expression under the sign of the integral is equal to

$$\frac{d}{dx} \left(\frac{(\lambda x)^n}{n!} e^{-\lambda x} \right).$$

Thus, we can compute the integral as follows:

$$\mathbb{P}[N_t = n] = \left(\frac{(\lambda x)^n}{n!} e^{-\lambda x} \right) \Big|_{x=0}^{x=t} = \frac{(\lambda t)^n}{n!} e^{-\lambda t},$$

Reference only

where the last step holds since we assumed that $n \neq 0$. □

REMARK 3.3.3. From the above theorem it follows that the renewal function of the Poisson process is given by $H(t) = \mathbb{E}N_t = \lambda t$.

For the next theorem let $U_1, \ldots, U_n$ be independent random variables which are uniformly distributed on the interval $[0, t]$. Denote by $U_{(1)} \leq \ldots \leq U_{(n)}$ the order statistics of $U_1, \ldots, U_n$.

THEOREM 3.3.4. *The conditional distribution of the random vector $(S_1, \ldots, S_n)$ given that $\{N_t = n\}$ coincides with the distribution of $(U_{(1)}, \ldots, U_{(n)})$:*

$$(S_1, \ldots, S_n)|\{N_t = n\} \stackrel{d}{=} (U_{(1)}, \ldots, U_{(n)}).$$

PROOF. We will compute the densities of both vectors and show these densities are equal.

STEP 1. The joint density of the random variables $(\xi_1, \ldots, \xi_{n+1})$ has (by independence) the product form

$$f_{\xi_1,\ldots,\xi_{n+1}}(u_1, \ldots, u_{n+1}) = \prod_{k=1}^{n+1} \lambda e^{-\lambda u_k}, \quad u_1, \ldots, u_{n+1} > 0.$$

STEP 2. We compute the joint density of $(S_1, \ldots, S_{n+1})$. Consider a linear transformation A defined by

$$A(u_1, u_2, \ldots, u_{n+1}) = (u_1, u_1 + u_2, \ldots, u_1 + \ldots + u_{n+1}).$$

The random variables $(S_1, \ldots, S_{n+1})$ can be obtained by applying the linear transformation A to the variables $(\xi_1, \ldots, \xi_{n+1})$:

$$(S_1, \ldots, S_{n+1}) = A(\xi_1, \ldots, \xi_{n+1}).$$

The determinant of the transformation A is 1 since the matrix of this transformation is triangular with 1's on the diagonal. By the density transformation theorem, the density of $(S_1, \ldots, S_{n+1})$ is given by

$$f_{S_1,\ldots,S_{n+1}}(t_1, \ldots, t_{n+1}) = \prod_{k=1}^{n+1} \lambda e^{-\lambda(t_k - t_{k-1})} = \lambda^{n+1} e^{-\lambda t_{n+1}},$$

where $0 = t_0 < t_1 < \ldots < t_{n+1}$. Otherwise, the density vanishes. Note that the formula for the density depends only on t_{n+1} and does not depend on $t_1, \ldots, t_n$.

STEP 3. We compute the conditional density of $(S_1, \ldots, S_n)$ given that $N_t = n$. Let $0 < t_1 < \ldots < t_n < t$. Intuitively, the conditional density of $(S_1, \ldots, S_n)$ given that $N_t = n$ is given by

$$\begin{aligned} f_{S_1,\ldots,S_n}(t_1, \ldots, t_n|N_t = n) &= \lim_{\varepsilon \downarrow 0} \frac{\mathbb{P}[t_1 < S_1 < t_1 + \varepsilon, \ldots, t_n < S_1 < t_n + \varepsilon | N_t = n]}{\varepsilon^n} \\ &= \lim_{\varepsilon \downarrow 0} \frac{\mathbb{P}[t_1 < S_1 < t_1 + \varepsilon, \ldots, t_n < S_n < t_n + \varepsilon, N_t = n]}{\varepsilon^n \mathbb{P}[N_t = n]} \\ &= \lim_{\varepsilon \downarrow 0} \frac{\mathbb{P}[t_1 < S_1 < t_1 + \varepsilon, \ldots, t_n < S_n < t_n + \varepsilon, S_{n+1} > t]}{\varepsilon^n \mathbb{P}[N_t = n]}. \end{aligned}$$



Reference only

Using the formula for the joint density of $(S_1, \ldots, S_{n+1})$ and noting that this density does not depend on $t_1, \ldots, t_n$, we obtain that

$$\frac{\mathbb{P}[t_1 < S_1 < t_1 + \varepsilon, \ldots, t_n < S_n < t_n + \varepsilon, S_{n+1} > t]}{\varepsilon^n \mathbb{P}[N_t = n]} = \frac{\int_t^\infty \lambda^{n+1} e^{-\lambda t_{n+1}} dt_{n+1}}{\mathbb{P}[N_t = n]} = \frac{n!}{t^n},$$

where in the last step we used that N_t has Poisson distribution with parameter λt. So, we have

$$f_{S_1,\ldots,S_n}(t_1, \ldots, t_n | N_t = n) = \begin{cases} \frac{n!}{t^n}, & \text{for } 0 < t_1 < \ldots < t_n < t, \\ 0, & \text{otherwise.} \end{cases}$$

Step 4. The joint density of the order statistics $(U_{(1)}, \ldots, U_{(n)})$ is known (Stochastik I) to be given by

$$f_{U_{(1)},\ldots,U_{(n)}}(t_1, \ldots, t_n) = \begin{cases} \frac{n!}{t^n}, & \text{for } 0 < t_1 < \ldots < t_n < t, \\ 0, & \text{otherwise.} \end{cases}$$

This coincides with the conditional density of $(S_1, \ldots, S_n)$ given that $N_t = n$, thus proving the theorem. □

Theorem 3.3.5. *The Poisson process $\{N_t : t \geq 0\}$ has independent increments and these increments have Poisson distribution, namely for all $t, s \geq 0$ we have*

$$N_{t+s} - N_t \sim \text{Poi}(\lambda s).$$

Proof. Take some points $0 = t_0 \leq t_1 \leq \ldots \leq t_n$. We determine the distribution of the random vector

$$(N_{t_1}, N_{t_2} - N_{t_1}, \ldots, N_{t_n} - N_{t_{n-1}}).$$

Take some $x_1, \ldots, x_n \in \mathbb{N}_0$. We compute the probability

$$P := \mathbb{P}[N_{t_1} = x_1, N_{t_2} - N_{t_1} = x_2, \ldots, N_{t_n} - N_{t_{n-1}} = x_n].$$

Let $x = x_1 + \ldots + x_n$. By definition of conditional probability,

$$P = \mathbb{P}[N_{t_1} = x_1, N_{t_2} - N_{t_1} = x_2, \ldots, N_{t_n} - N_{t_{n-1}} = x_n | N_{t_n} = x] \cdot \mathbb{P}[N_{t_n} = x].$$

Given that $N_{t_n} = x$, the Poisson process has x renewals in the interval $[0, t_n]$ and by Theorem 3.3.4 these renewals have the same distribution as x independent random variables which have uniform distribution on the interval $[0, t_n]$, after arranging them in an increasing order. Hence, in order to compute the conditional probability we can use the multinomial distribution:

$$P = \left(\frac{x!}{x_1! \ldots x_n!} \prod_{k=1}^n \frac{(t_k - t_{k-1})^{x_k}}{t_n^{x_k}} \right) \cdot \frac{(\lambda t_n)^x}{x!} e^{-\lambda t_n}.$$

After making transformations we arrive at

$$P = \prod_{k=1}^n \left(\frac{(\lambda(t_k - t_{k-1}))^{x_k}}{x_k!} e^{-\lambda(t_k - t_{k-1})} \right).$$

From this formula we see that the random variables N_{t_1}, $N_{t_2} - N_{t_1}, \ldots$, $N_{t_n} - N_{t_{n-1}}$ are independent and that they are Poisson distributed, namely

$$N_{t_k} - N_{t_{k-1}} \sim \text{Poi}(\lambda(t_k - t_{k-1})).$$

This proves the theorem. □

Reference only

THEOREM 3.3.6. *The Poisson process has stationary increments.*

PROOF. Take some $h \geq 0$, and some $0 \leq t_1 \leq t_2 \leq \ldots \leq t_n$. We have to show that the distribution of the random vector

$$(N_{t_1+h} - N_{t_0+h}, N_{t_2+h} - N_{t_1+h}, \ldots, N_{t_n+h} - N_{t_{n-1}+h})$$

does not depend on h. However, we know from Theorem 3.3.5 that the components of this vector are independent and that

$$N_{t_k+h} - N_{t_{k-1}+h} \sim \text{Poi}(\lambda(t_k - t_{k-1})),$$

which does not depend on h. □

N(t) is a Poisson process

Poisson Thinning Theorem

- **Proposition 5.3 (Poisson Thinning Theorem)** If the classification probability is time-dependent, and the event arriving at time s will be classified as type i with probability $P_i(s)$. Then, $N_i(t)$, $i = 1, ..., k$, are independent Poisson random variables having means

$$E[N_i(t)] = \lambda \int_0^t P_i(s)ds.$$

$N_i(t)$ = # of type i in $[0, t]$, $1 \le i \le k$.

Proof. Given $N(t)=n$, according to the previous thm. the n arrival times $T_1,\cdots,T_n$ are order statistics of n iid uniform on $[0,t]$. In particular, The time T of one event occurs $\sim$ Uniform on $[0,T]$.

$$P_i \overset{\text{def}}{=} P(\text{the event is of type } i)$$

$$= EP(\text{the event is of type } i \mid T)$$

$$= E\, P_i(T)$$

$$= \frac{1}{t}\int_0^t P_i(s)\,ds.$$

Obviously, $P_1+\cdots+P_k=1$. Thus

$$P(N_1(t)=n_1, N_2(t)=n_2,\cdots, N_k(t)=n_k)$$

$$= P(N_1(t)=n_1, N_2(t)=n_2,\cdots, N_k(t)=n_k \mid N(t)=n_1+\cdots+n_k) \cdot P(N(t)=n_1+\cdots+n_k)$$

This is the same as rolling a k-side die $n_1+\cdots+n_k$ times, where $P(i\text{-dot})=P_i$. Then $(N_1(t),\cdots,N_k(t))$ is multinomial dist. Thus the above probab is

$$\binom{n_1+\cdots+n_k}{n_1,n_2,\cdots,n_k} P_1^{n_1}\cdots P_k^{n_k} \cdot \frac{(\lambda t)^{n_1+\cdots+n_k}}{(n_1+\cdots+n_k)!} e^{-\lambda t}$$

$$= \frac{(\lambda p_1 t)^{n_1}}{n_1!} \cdots \frac{(\lambda p_k t)^{n_k}}{n_k!}$$

$$= P(Z_1 = n_1) \cdot P(Z_2 = n_2) \cdots P(Z_k = n_k)$$

where $Z_i \sim Poi(\lambda p_i t)$. Notice

$$\lambda p_i t = \lambda \int_0^t p_i(s)\,ds.$$

Eg. $k = 6$, $t = 1$.

$$p_1(s) = p_2(s) = p_3(s) = p_4(s) = p_5(s) = \frac{1-s}{6}.$$

$$p_6(s) = \frac{1+5s}{6}, \quad 0 \le s \le 1.$$

$$p_i = \int_0^1 p_1(s)\,ds = \int_0^1 \frac{1-s}{6}\,ds = \frac{1}{12}$$

$i = 1, 2, 3, 4, 5.$

$$p_6(s) = \int_0^1 p_6(s)\,ds = \int_0^1 \frac{1+5s}{6}\,ds = \frac{1}{6} + \frac{5}{12} = \frac{7}{12}$$

Randomly select a time in $[0,1]$ and roll a die. what is the chance it is 1-dot, 2-dot, ···, 6-dot?

In fact, $P(1\text{-dot}) = E\,P(1\text{-dot} \mid T)$, where T has uniform dist. on $[0,1]$. The above $= E\,P_1(T)$

$$= \int_0^1 p_1(t) \cdot g(t)\,dt = \int_0^1 p_1(t)\,dt, \text{ where } g(t) \text{ is the pdf of } T.$$

Example

- **Example 5.18 (An Infinite Server Queue)** Suppose that customers arrive at a service station in accordance with a Poisson process with rate λ. Upon arrival the customer is immediately served by one of an infinite number of possible servers, and the service times are assumed to be independent with a common distribution G. What is the distribution of $X(t)$, the number of customers that have completed service by time t? What is the distribution of $Y(t)$, the number of customers that are being served at time t? What is the joint distribution of $(Y(t), Y(t+s))$?

Solution. A customer C comes at time s, C belongs to Type I, if the service time g satisfies $g+s \le t$; Otherwise, C belongs to type II.

Chance of type I : $P_1(s) = P(g+s \le t) = G(t-s)$,

- - - ~ - - II : $P_2(s) = P(g+s > t) = 1 - G(t-s)$

for $0 \le s < t$.

According to Poisson thinning thm, $X(t)$ and $Y(t)$ are indept r.v.'s. Also $X(t) \sim Poi(\lambda_1)$, $Y(t) \sim Poi(\lambda_2)$ with

$$\lambda_1 = \lambda \int_0^t G(t-s)\,ds = \lambda \int_0^t G(s)\,ds;$$

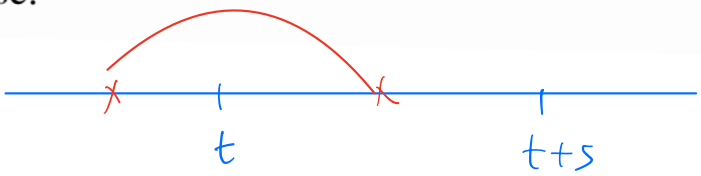
$$\lambda_2 = \lambda \int_0^t [1 - G(t-s)]\,ds = \lambda - \lambda \int_0^t G(s)\,ds.$$

Now we compute joint dist. of $Y(t)$ and $Y(t+s)$.

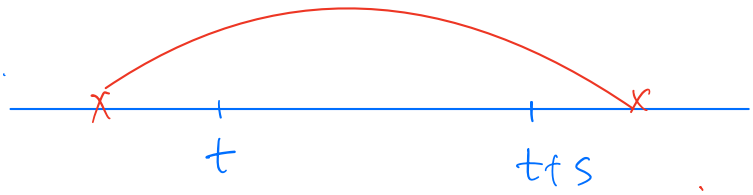
At time $t+s$ we look back and classify customers.

type 1: if he arrives before time t and completes service between t and $t+s$,
type 2: if he arrives before t and completes service after $t+s$,
type 3: if he arrives between t and $t+s$ and completes service after $t+s$,
type 4: otherwise.

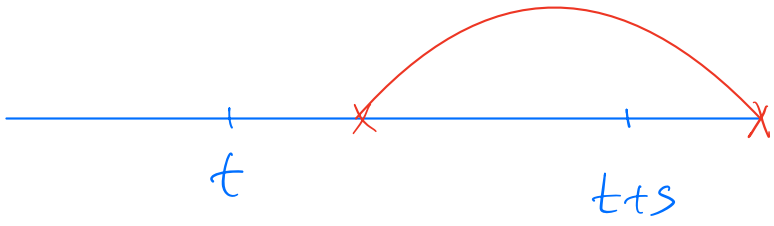
type 1 :



type 2 :



type 3:



Let $P_i(y)$ be the probab. customer C belong to type i when C comes at time y. Let g = serive time. g has cdf $G(x)$.

$$P_1(y) = P(t < g+y < t+s) = G(t+s-y) - G(t-y), \quad y<t,$$

$$P_2(y) = P(g+y > t+s) = 1 - G(t+s-y), \quad y<t,$$

$$P_3(y) = P(g+y > t+s) = 1 - G(t+s-y), \quad t<y<t+s,$$

$$P_4(y) = 1 - P_1(y) - P_2(y) - P_3(y) - P_4(y)$$

Let $N_i = N_i(s+t)$ denote the number of type i events occur by $s+t$ Then N_1, N_2, N_3, N_4 are indept $Poi(\lambda_i)$ with

$$\lambda_i = \lambda \int_0^{t+s} P_i(y)\,dy, \quad i = 1,2,3,4.$$

Notice

$$Y(t) = N_1 + N_2, \quad Y(t+s) = N_2 + N_3.$$

We then know everything and anything on joint dist. of $(Y(t), Y(t+s))$.

Non-homogeneous Poisson Process

- The counting process $\{N(t), t \geq 0\}$ is said to be a nonhomogeneous Poisson process with intensity function $\lambda(t)$, $t \geq 0$, if
  1. $N(0) = 0$.
  2. $\{N(t), t \geq 0\}$ has independent increments.
  3. $P(N(t+h) - N(t) \geq 2) = o(h)$.
  4. $P(N(t+h) - N(t) = 1) = \lambda(t)h + o(h)$.

- **Theorem 5.3** If $\{N(t) : t \geq 0\}$ is a non-homogeneous Poisson process with rate $\lambda(t)$, then $N(t+s) - N(s)$ follows a Poisson distribution with mean equal to

$$m_s(t) \triangleq \int_s^{t+s} \lambda(u)du.$$

Non-homogeneous Poisson Process

- **Poisson Thinning Theorem:** If the classification probability is time-dependent, and the event arriving at time t will be classified as type i with probability $P_i(s)$. Then, $N_i(t)$, $i = 1, ..., k$, are independent Poisson random variables having means

$$E[N_i(t)] = \int_0^t \lambda(s) P_i(s) ds$$

- **Conditional Distribution of Arrival Times:** Given that $N(t) = n$, the n arrival times $S_1, ..., S_n$ have the same distribution as the *order statistics* corresponding to n independent random variables distributed on the interval $(0, t)$ with probability density function

$$f(x) = \frac{\lambda(x)}{\int_0^t \lambda(s) ds}, 0 < x < t.$$

Application Example

- **Sampling Algorithm** Simulate a non-homogeneous Poisson process with rate $\lambda(t)$ on $[0, T]$

Method 1. Suppose $\lambda(t) \leq \lambda$ for all $t \in [0, T]$. First generate a Poisson process with rate λ. Let $0 < S_1 < S_2 < ... < S_n < T$ be the arrival times. For each i, generate a Bernoulli random variable I_i with $p = \lambda(S_i)/\lambda$. Output $\{S_i : I_i = 1, 1 \leq i \leq n\}$.

Method 2. Let $m = \int_0^T \lambda(t)dt$. Simulate a Poisson random variable $N$with mean m. Given N, simulate i.i.d. random variables with density

$$f(t) = \frac{\lambda(t)}{m}, 0 < t < T.$$

Examples

- **Example 5.24** Siegbert runs a hot dog stand that opens at 8 A.M. From 8 until 11 A.M. customers seem to arrive, on the average, at a steadily increasing rate that starts with an initial rate of 5 customers per hour at 8 A.M. and reaches a maximum of 20 customers per hour at 11 A.M. From 11 A.M. until 1 P.M. the (average) rate seems to remain constant at 20 customers per hour. However, the (average) arrival rate then drops steadily from 1 P.M. until closing time at 5 P.M. at which time it has the value of 12 customers per hour. If we assume that the numbers of customers arriving at Siegbert's stand during disjoint time periods are independent, then what is a good probability model for the preceding? What is the probability that no customers arrive between 8:30 A.M. and 9:30 A.M. on Monday morning? What is the expected number of arrivals in this period?

Solution. We propose a model with Poisson process with non-homogeneity. One can see the rate

$$\lambda(t) = \begin{cases} 0, & 0 \le t \le 8; \\ 5+5(t-8), & 8 < t \le 11; \\ 20, & 11 < t \le 13; \\ 20-2(t-13), & 13 < t \le 17; \\ 0, & 17 \le t \le 24, \end{cases}$$

and $\lambda(t) = \lambda(t')$, if $t > 24$,

where t' = reminder after t divided by 24.

Use $N(t) - N(s) \sim \text{Poi}\left(\int_s^t \lambda(u)du\right)$

- $P(N(9.5) - N(8.5) = 0)$

$$= \exp\left(-\int_{8.5}^{9.5} \lambda(s)ds\right)$$

Now $\int_{8.5}^{9.5} \lambda(s)ds = \int_{0.5}^{1.5} (5+5u)du = 10$

Thus, the above chance $= e^{-10}$.

- $E[N(9.5) - N(8.5)] = E\,\text{Poi}(10) = 10.$

Examples

- **Example 5.25 (The Output Process of an Infinite Server Poisson Queue)** It turns out that the output process of the $M/G/\infty$ queue, that is, of the infinite server queue having Poisson arrivals and general service distribution G, is a non-homogeneous Poisson process having intensity function $\lambda(t) = \lambda G(t)$.

Markovian General Infinite queue

Check textbook for reasoning

Compound Poisson Process

- A stochastic process $\{X(t), t \geq 0\}$ is said to be a compound Poisson process if it can be represented as

$$X(t) = \sum_{i=1}^{N(t)} Y_i,$$

where $\{N(t), t \geq 0\}$ is a Poisson process, and $\{Y_i, i \geq 1\}$ is a family of independent and identically distributed random variables that is also independent of $\{N(t), t \geq 0\}$.

- What is the value of $E[X(t)]$ and Var$(X(t))$?

Solution. Set $\mu = EX_1$, $\sigma^2 = \mathrm{Var}(X_1)$.

As seen earlier,

$$EX(t) = E\,E[X(t) \mid N(t)]$$

$$= E\,N(t) \cdot EY_1 = \mu \cdot \lambda t$$

$$\mathrm{Var}(X(t)) = E\,\mathrm{Var}(X(t) \mid N(t)) + \mathrm{Var}(E(X(t) \mid N(t)))$$

$$= E[\sigma^2 N(t)] + \mathrm{Var}(N(t) \cdot \mu)$$

$$= \lambda t \sigma^2 + \mu^2 (\lambda t)$$

$$= \lambda t \cdot E(X_i^2)$$

Since $N(t) \sim \mathrm{Poi}(\lambda t)$.

Examples

- **Example 5.26** Suppose that families migrate to an area at a Poisson rate $\lambda = 2$ per week. If the number of people in each family is independent and takes on the values 1, 2, 3, 4 with respective probabilities 1/6 , 1/3 , 1/3 , 1/6, then what is the expected value and variance of the number of individuals migrating to this area during a fixed five-week period?

Solution. Model $X(t) = \sum_{i=1}^{N(t)} Y_i$

where $N(t)$ is a Poisson process with rate $\lambda = 2$, and $Y_1, Y_2, \cdots$ are iid r.v.'s with dist.

x	1	2	3	4
$p(x)$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{6}$

$$EX = 1\cdot\frac{1}{6} + 2\cdot\frac{1}{3} + 3\cdot\frac{1}{3} + 4\cdot\frac{1}{6} = \frac{5}{2}$$

$$EX^2 = 1\cdot\frac{1}{6} + 2^2\cdot\frac{1}{3} + 3^2\cdot\frac{1}{3} + 4^2\cdot\frac{1}{6} = \frac{43}{6}$$

Thus, $E\,X(5) = (5\cdot 2)\cdot\frac{5}{2} = 25$,

$$\mathrm{Var}(X(5)) = (5\cdot 2)\frac{43}{3} = \frac{215}{3}.$$

Chapter Review

- Exponential distribution
  - Memoryless property
  - Minimum of independent exponential r.v.'s
- Poisson process
  - Counting process; independent / stationary increments
  - 4 axioms of Poisson process
  - Derivation of inter-arrival time / increment distribution
  - Poisson thinning theorem
- Generalization of Poisson process
  - Non-homogeneous Poisson process
  - Compound Poisson process